

Standard Model of Elementary Particles

three generations of matter
(fermions)

interactions / force carriers
(bosons)

I

II

III

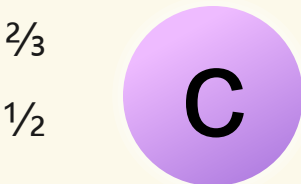
mass
charge
spin

$\approx 2.16 \text{ MeV}/c^2$



up

$\approx 1.273 \text{ GeV}/c^2$



charm

$\approx 172.57 \text{ GeV}/c^2$



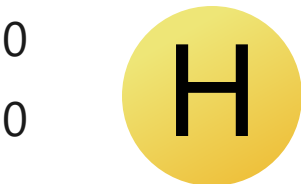
top

0



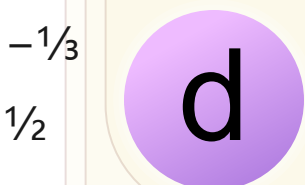
gluon

$\approx 125.2 \text{ GeV}/c^2$



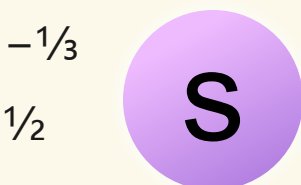
higgs

$\approx 4.7 \text{ MeV}/c^2$



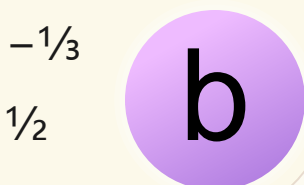
down

$\approx 93.5 \text{ MeV}/c^2$



strange

$\approx 4.183 \text{ GeV}/c^2$



bottom

0



photon

$\approx 0.511 \text{ MeV}/c^2$



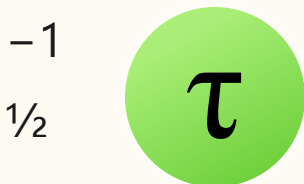
electron

$\approx 105.66 \text{ MeV}/c^2$



muon

$\approx 1.77693 \text{ GeV}/c^2$



tau

$\approx 91.188 \text{ GeV}/c^2$



Z boson

$< 0.8 \text{ eV}/c^2$



electron
neutrino

$< 0.17 \text{ MeV}/c^2$



muon
neutrino

$< 18.2 \text{ MeV}/c^2$



tau
neutrino

$\approx 80.3692 \text{ GeV}/c^2$



W boson

QUARKS

LEPTONS

GAUGE BOSONS
VECTOR BOSONS

SCALAR BOSONS